STAT 1210 - Quiz 3 Grading Guide

Part III: Foundations of Statistical Inference

Fall 2026

Marking Notes

- The quiz is out of 20 marks: Question 1 (7), Question 2 (7), Question 3 (6).
- Half marks may be awarded as indicated in each part.
- Accept small rounding differences (for example, z-table values versus exact software values).
- Follow-through: if a student makes an early numerical error but uses the correct method afterwards, award the later method marks.
- Explanations must refer to the context of the question to receive full marks.
- Key answers are listed for quick reference; see the Solutions document for full working.

	Part 1	Part 2	Part 3	Part 4	Part 5	Total
Question 1	1	1.5	2	1.5	1	7
Question 2	1	1.5	1	2.5	1	7
Question 3	2	1	1	1	1	6
						20

Question 1: Wireworm Damage in PEI Potato Fields (7 marks)

Part 1 (1 mark). *Key:* all 4000 registered PEI potato fields; categorical.

- 0.5 for the population (all 4000 registered PEI potato fields).
- 0.5 for identifying the variable as categorical.

Part 2 (1.5 marks). *Key:* A convenience, B stratified (proportional), C SRS; $0.45, 0.30, 0.25$ in both.

- 0.25 for each correct design (0.75 total).
- 0.75 for showing the population and sample proportions (or equivalently $200 \times 0.45 = 90$, etc.) and stating that they match.

Part 3 (2 marks). *Key:* $\mu_{\hat{p}} = 0.15$, $\sigma_{\hat{p}} \approx 0.025$; $np = 30$, $n(1-p) = 170$; $200 \leq 400$.

- 0.5 for the mean $\mu_{\hat{p}} = 0.15$.
- 0.5 for the standard deviation ≈ 0.025.
- 0.5 for the expected count condition, showing both $np = 30$ and $n(1-p) = 170$ are at least 10. Award 0.25 if only one is checked.
- 0.5 for the 10% condition, showing $200 \leq 400$.

Part 4 (1.5 marks). *Key:* $z \approx 1.98$, $P \approx 0.0239$.

- 0.5 for the correct z-score ≈ 1.98.
- 0.5 for the probability ≈ 0.024 (accept 0.0238 to 0.0239). Award 0.25 if $P(Z \leq 1.98) = 0.9761$ is reported instead of the upper tail.
- 0.5 for a correct contextual interpretation referring to repeated simple random samples of 200 fields when $p = 0.15$.

Part 5 (1 mark). *Key:* observational study.

- 0.5 for identifying it as an observational study.
- 0.5 for the reason: no treatment (rotation) was imposed or randomly assigned by the researchers.

Question 2: Shell Length of Cultured Mussels in Malpeque Bay (7 marks)

Part 1 (1 mark). *Key:* μ = population mean shell length; $SE \approx 0.857$ mm.

- 0.5 for defining μ as a **population** mean in context (not the sample mean).
- 0.5 for the standard error ≈ 0.857 mm.

Part 2 (1.5 marks). *Key:* $m = 1.680$; $(56.420,\ 59.780)$ mm.

- 0.5 for the correct formula $\bar{x} \pm z^*\sigma/\sqrt{n}$.
- 0.5 for the margin of error ≈ 1.680.
- 0.5 for the correct interval endpoints (accept 56.42 and 59.78, or 56.4 and 59.8).

Part 3 (1 mark).

- 0.5 for a statement about the **population mean** shell length, in context, with the endpoints and units.
- 0.5 for a correct meaning of "95% confident" (refers to the long-run success rate of the method). Award 0 for this half if the student says there is a 95% probability that μ is in this particular interval, or that 95% of individual mussels fall in the interval.

Part 4 (2.5 marks). *Key:* $H_0 : \mu = 60$ vs $H_a : \mu \neq 60$; $z \approx -2.22$; p-value ≈ 0.0264; reject H_0.

- 0.5 for correct hypotheses about μ (two-sided). Deduct 0.25 if written in terms of $\bar{x}$.
- 0.5 for the test statistic $z \approx -2.22$.
- 0.75 for the two-sided p-value ≈ 0.0264. Award 0.25 if the one-sided value 0.0132 is given.
- 0.25 for the correct decision (reject H_0) based on comparing the p-value with α.
- 0.5 for a conclusion in context (accept "does not prove" style language if correct).

Part 5 (1 mark). *Key:* 60 is outside $(56.420,\ 59.780)$, so reject.

- 0.5 for noting that 60 lies outside the 95% interval.
- 0.5 for linking this to the test: a 95% interval matches a two-sided test at $\alpha = 0.05$, so a null value outside the interval is rejected, the same decision as Part 4.

Question 3: Emergency Department Wait Times on PEI (6 marks)

Part 1 (2 marks). *Key:* (a) $96.04 \to 97$; (b) $165.894 \to 166$.

- 0.5 for the correct formula $n = (z^*\sigma/m)^2$ or equivalent setup.
- 0.5 for 96.04 in (a) and 0.5 for 165.894 in (b).
- 0.5 for rounding **up** to $n = 97$ and $n = 166$. Award 0 for this item if the student rounds down (96, 165).

Part 2 (1 mark). *Key:* $384.16 \to 385$; about 4 times Part 1(a).

- 0.5 for $n = 385$.
- 0.5 for the comparison: about 4 times the sample size in Part 1(a) (explanation via the square-root relationship is encouraged but not required).

Part 3 (1 mark). *Key:* undercoverage / selection bias.

- 1 for naming undercoverage or selection bias (a brief justification is welcome but not required, since the question asks only to name the error).
- 0.5 for naming a valid non-sampling error not tied to the weekday daytime restriction (for example, nonresponse or measurement error in recorded arrival times).

Part 4 (1 mark).

- 0.5 for explaining statistical significance (small p-value; the large sample makes even a small effect detectable).
- 0.5 for explaining that practical importance concerns the size of the effect, with a contextual judgement that 1.2 minutes is small.

Part 5 (1 mark). *Key:* $1 - (0.95)^{10} \approx 0.401$.

- 0.5 for the correct complement-rule setup $1 - (0.95)^{10}$.

- 0.5 for the value ≈ 0.401.